

Design and Development Task in Control Systems (EE347)

Project End Semester-Report on

Hybrid DC–DC Converter Fed 5-Level CHB Multilevel Inverter System

Submitted in partial fulfilment of the requirements for the degree of

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in

ELECTRICAL AND ELECTRONICS ENGINEERING

by

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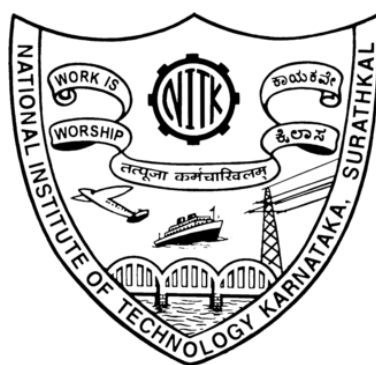
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DECLARATION

We hereby declare that the report entitled “**Hybrid DC–DC Converter Fed 5-Level CHB Multilevel Inverter System**” is a Bonafide record of the project work carried out by us in fulfilment of the requirements of the course **EE347 - Design and Development Task in Control Systems** in the Department of Electrical and Electronics Engineering, National Institute of Technology Karnataka, Surathkal. The work presented in this report is original and has not been submitted to any other University or Institution for the award of any degree, diploma, or certification.

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ABSTRACT

This mini project focuses on the design, working, and Simulink implementation of two basic DC-DC converters, namely a boost converter and a buck converter, operating from a 10 V DC source at a switching frequency of 50 kHz. The project also provides a brief overview of a 5-level H-bridge multilevel inverter that can be used as the final AC conversion stage in a power electronic system. The boost converter and buck converter component values are not treated as fixed givens; instead, the inductor and capacitor are selected by applying design considerations such as continuous conduction mode, ripple current, and output voltage ripple.

The boost converter uses a selected inductor of 470 microhenry and a capacitor of 220 microfarad after design verification, while the buck converter uses a selected inductor of 2.2 millihenry and a capacitor of 100 microfarad based on the same design approach. The report explains the working principle of each converter, derives the important design formulas, calculates practical component values, and lists the steps needed to build the simulations in MATLAB/Simulink.

A short conceptual discussion of the 5-level inverter is included to connect the DC-DC stage to the AC output stage. The report is written in a format suitable for a mini project submission and can be expanded with screenshots of Simulink models, waveforms, and hardware photographs.

Keywords: SPWM, Cascaded H-Bridge (CHB), Multi-Level Inverter, MATLAB/Simulink, THD, LC Filter, Power Quality

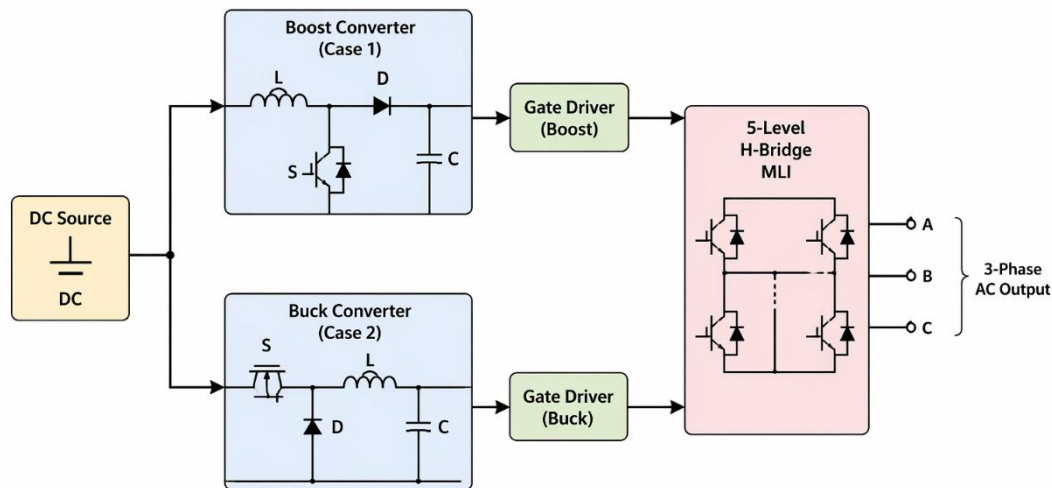
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INTRODUCTION

1.1 Background

Power electronic converters are used to regulate and condition electrical power so that a source can supply a load with the required voltage, current, and waveform. In modern applications, a low-voltage DC source such as a battery, solar panel, or laboratory supply often needs to be converted into a regulated DC bus and then into AC. This is commonly achieved by using a DC-DC converter stage followed by an inverter stage.



1.2 Scope

In this project, the first stage is a DC-DC converter designed in two cases:

1. Boost Converter Case: Input DC voltage is stepped up from 10 V to 20 V
2. Buck Converter Case: Input DC voltage is stepped down from 10 V to 5V

Both converters operate from the same 10 V DC input source and use a switching frequency of 50 kHz. The project also includes a brief overview of a 5-level H-bridge multilevel inverter, which is the final stage used to synthesize AC voltage with improved waveform quality.

1.3 Objectives

The purpose of this report is to present:

- The working principle of a boost converter
- Design formulas and calculations for boost converter component selection
- Practical component values for boost converter simulation and hardware
- The working principle of a buck converter
- Design formulas and calculations for buck converter component selection
- Practical component values for buck converter simulation and hardware
- A conceptual explanation of 5-level multilevel inverter topology
- Complete Simulink implementation methodology

1.4 Design Philosophy

A critical technical point throughout this report is that the numerical design values are obtained through systematic analysis, not assumed as fixed inputs. The duty ratio and switching frequency are given, but:

- The output voltage is calculated from the duty ratio relation
- The output current is calculated from the output voltage and load resistance
- The inductor value is derived from assuming an acceptable current ripple and applying the ripple formula
- The capacitor value is derived from assuming an acceptable output voltage ripple
- Both components are then verified to satisfy continuous conduction mode requirements

This approach ensures engineering rigor and practical component selection suitable for both simulation and hardware implementation.

Voltage and Load Specifications:

For reference:

- 10 V source boosted only to 20 V DC is not sufficient to directly produce 230 V_{rms} AC
- A practical 230 V_{rms} system would require a higher DC-link voltage or additional boost stages

SYSTEM OVERVIEW

The complete power conversion chain can be described as follows:

2.1 Boost Converter Path

Parameter	Value
Input source	10 V DC
Converter	Boost converter
Switching frequency	50 kHz
Duty ratio	0.5
Load resistance	51 ohm
Inductor L	100 μH (design-selected)
Output capacitor C	220 μF (design-selected)
Output voltage (ideal)	20 V DC

2.2 Buck Converter Path

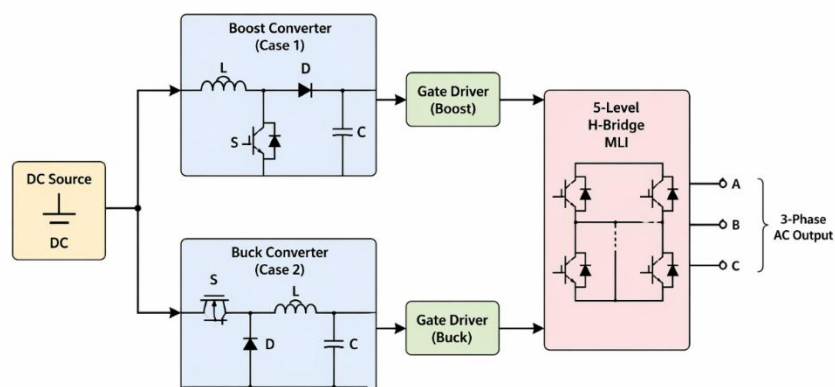
Parameter	Value
Input source	10 V DC
Converter	Buck converter
Switching frequency	50 kHz
Duty ratio	0.5
Load resistance	51 ohm
Inductor L	2.2 mH (design-selected)
Output capacitor C	100 μ F (design-selected)
Output voltage (ideal)	5 V DC

Final Integration Stage: 5-Level Multilevel Inverter

The inverter provides the final AC conversion:

- Generates a stepped AC waveform from the regulated DC bus
- Uses PWM-based gate driver signals for switching control
- Reduces harmonics compared to conventional two-level inverters
- Produces improved AC output waveform quality

2.3 System Block Diagram



2.4 Functional Flow

This arrangement is common in:

- Renewable energy systems (solar, wind)
- Battery-fed inverters and UPS systems
- Power conditioning and DC microgrids
- Electric vehicle auxiliary power stages
- Laboratory and educational platforms

The DC-DC converter stage either steps up voltage (boost) or steps down voltage (buck) depending on the application requirement, creating an intermediate regulated DC bus. The inverter stage then converts this DC bus to AC with improved harmonic performance.

BOOST CONVERTER DESIGN

The boost converter is a fundamental DC-DC converter topology that provides step-up voltage conversion. This section presents the working principle, mathematical foundations, design methodology, and Simulink implementation for a 10 V to 20 V converter at 50 kHz switching frequency.

3.1 Working Principle

A boost converter is a step-up DC-DC converter that increases the output voltage above the input voltage using five key components: inductor, switch (MOSFET), freewheeling diode, capacitor, and load. The converter operates in two distinct intervals during each switching cycle.

Switch ON Interval

When the MOSFET is ON:

- The inductor is connected directly across the input source
- The inductor current increases linearly as it stores energy
- The diode is reverse-biased (blocks current)
- The load is supplied primarily by the output capacitor, which discharges
- The output voltage temporarily drops by a small amount

During this interval, the inductor stores energy in its magnetic field according to:

$$V_L = L \frac{di_L}{dt}$$

Switch OFF Interval

When the MOSFET is OFF:

- The inductor current cannot change instantaneously
- The inductor polarity reverses to maintain current continuity
- The diode becomes forward-biased and conducts
- The inductor releases its stored energy to the output capacitor and load
- The output capacitor charges to higher voltage

The inductor voltage adds to the source voltage during the OFF interval, resulting in an output voltage greater than the input voltage. This is the fundamental step-up mechanism.

Energy Transfer Mechanism

Energy stored in inductor during ON interval:

$$E_L = \frac{1}{2} L I_L^2$$

This energy is transferred to the load and capacitor during the OFF interval, creating a net energy flow path from input to output through the magnetic field.

3.2 Important Boost Converter Formulae

The following relationships apply for ideal continuous conduction mode (CCM) operation:

Output Voltage Relation

$$V_{out} = \frac{V_{in}}{1 - D}$$

where D is the duty ratio (fraction of time switch remains ON).

Duty Ratio Relation

$$D = 1 - \frac{V_{in}}{V_{out}}$$

Average Inductor Current

$$I_{L,avg} = \frac{I_{out}}{1 - D}$$

This shows that the inductor must carry more current than the output current to account for energy transfer.

Inductor Ripple Current

$$\Delta I_L = \frac{V_{in} \cdot D}{L \cdot f_s}$$

where:

- V_{in} = input voltage
- D = duty ratio
- L = inductor value
- f_s = switching frequency

Rearranging for inductor design:

$$L = \frac{V_{in} \cdot D}{\Delta I_L \cdot f_s}$$

Output Voltage Ripple

$$\Delta V_o = \frac{I_{out} \cdot D}{C \cdot f_s}$$

Rearranging for capacitor design:

$$C = \frac{I_{out} \cdot D}{\Delta V_o \cdot f_s}$$

Critical Inductance (CCM Boundary)

$$L_{crit} = \frac{D(1 - D)^2 R}{2f_s}$$

For continuous conduction mode, the selected inductor must satisfy: $L > L_{crit}$

Minimum Capacitance for Ripple Specification

$$C_{min} = \frac{I_{out} \cdot D}{\Delta V_o \cdot f_s}$$

3.3 Numerical Design for Boost Converter (Design-Based Component Selection)

This section demonstrates the complete step-by-step derivation of inductor and capacitor values from first principles.

Given Specifications

Parameter	Value
Input voltage	$V_{in} = 10 \text{ V}$
Duty ratio	$D = 0.5$
Switching frequency	$f_s = 50 \text{ kHz}$
Load resistance	$R = 51 \text{ ohm}$

Step 1: Calculate Output Voltage

Using the ideal boost relation:

$$V_{out} = \frac{V_{in}}{1 - D} = \frac{10}{1 - 0.5} = \frac{10}{0.5} = 20 \text{ V}$$

Result: Output voltage is 20 V

Step 2: Calculate Output Current

With the load resistance known:

$$I_{out} = \frac{V_{out}}{R} = \frac{20}{51} = 0.392 \text{ A}$$

Result: Output current is 0.392 A

Step 3: Assume Allowable Inductor Current Ripple and Calculate Inductor

Design Choice: For proper continuous conduction mode operation at 50 kHz with reasonable switch stress, assume a permissible inductor ripple current of 1 A.

$$L = \frac{V_{in} \times D}{\Delta I_L \times f_s} = \frac{10 \times 0.5}{1 \times 50 \times 10^3} = \frac{5}{50,000} = 100 \text{ } \mu\text{H}$$

Result: Calculated inductor value is 100 μH

Component Selection: Choose $L = 100 \text{ } \mu\text{H}$ (standard value available in commercial catalogues)

Step 4: Assume Allowable Output Voltage Ripple and Calculate Capacitor

Design Choice: For a well-regulated DC output with minimal voltage ripple suitable for powering sensitive circuits, assume output voltage ripple of 18 mV.

$$C = \frac{I_{out} \times D}{\Delta V_o \times f_s} = \frac{0.392 \times 0.5}{0.018 \times 50 \times 10^3} = \frac{0.196}{900} \approx 218 \text{ } \mu\text{F}$$

Result: Calculated capacitor value is approximately 218 μF

Component Selection: Choose $C = 220 \mu\text{F}$ (nearest standard electrolytic capacitor)

Final Design Summary

Component	Calculated Value	Selected Standard Value	Rating
Inductor L	100 μH	100 μH	50 V, 2 A
Capacitor C	218 μF	220 μF	35 V, 105 C

These values ensure continuous conduction mode operation and low output voltage ripple suitable for both simulation and hardware prototype development.

3.4 Simulation Steps in Simulink

The boost converter can be modelled in MATLAB/Simulink using the following systematic procedure:

Step 1: Project Setup

1. Open MATLAB and launch Simulink
2. Create a new blank model
3. Save it with filename: Boost_Converter.slx

Step 2: Add Power Source Components

1. Drag a DC Voltage Source block from the Simscape Electrical library into the model
 - Double-click to configure: Voltage = 10 V
2. Add a Ground block and connect it to the negative terminal

Step 3: Add Switch and Diode

1. Insert an N-Channel MOSFET (or ideal switch) block
 - Set initial parameters: $R_{on} = 0.01 \text{ ohm}$ (ideal switch behavior)
2. Add a Diode block in parallel with the switch (freewheeling path)
 - Configure as ideal diode
3. Connect the gate terminal to a PWM signal (to be added later)

Step 4: Add Energy Storage Components

1. Insert an Inductor block in series with the switch
 - Set inductance value = 100 microhenry (from design calculation)
2. Insert a Capacitor block on the output terminals
 - Set capacitance value = 220 microfarad (from design calculation)

Step 5: Add Load and Measurement Blocks

1. Add a Resistor block with resistance = 51 ohm as the load
2. Insert Voltage Measurement blocks to monitor:
 - Input voltage across DC source
 - Output voltage across the capacitor
3. Insert Current Measurement blocks to observe:
 - Inductor current
 - Switch current
 - Diode current

Step 6: Add PWM Control Signal

1. Drag in a Pulse Generator block or PWM Generator
 - Frequency = 50 kHz
 - Pulse width (duty cycle) = 50%
 - Period = 20 microseconds
 - Amplitude: 0 to 1 (or 0 to 5 V depending on MOSFET block requirements)

2. Connect this signal to the MOSFET gate terminal

Step 7: Add Measurement and Display Blocks

1. Insert Scope blocks to observe:
 - Gate pulse (PWM signal)
 - Inductor current waveform
 - Output voltage waveform
 - Diode current
2. Connect all signals appropriately

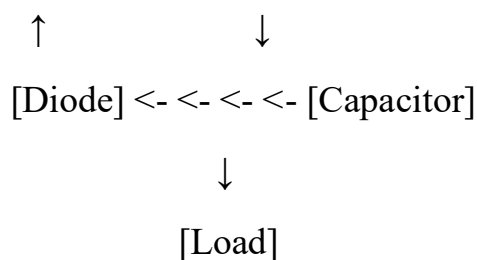
Step 8: Configure Simulation Parameters

1. Open Simulation > Configuration Parameters
2. Set solver parameters:
 - Solver: Fixed-step discrete
 - Step size: 1e-6 s (1 microsecond, adequate for 50 kHz simulation)
3. Set stop time: 20 ms (enough for 1000 switching cycles to reach steady state)

Step 9: Connect All Blocks in Standard Boost Topology

The basic circuit connection should be:

[DC Source] -> [Switch] -> [Inductor] -> [Output]



Step 10: Run Simulation

1. Press the Play button or press Ctrl+T
2. Run the simulation for 20 milliseconds
3. Open the Scope windows to observe the waveforms

4. Wait for signals to settle to steady-state

Step 11: Steady-State Verification

After settling time (typically after 5-10 ms):

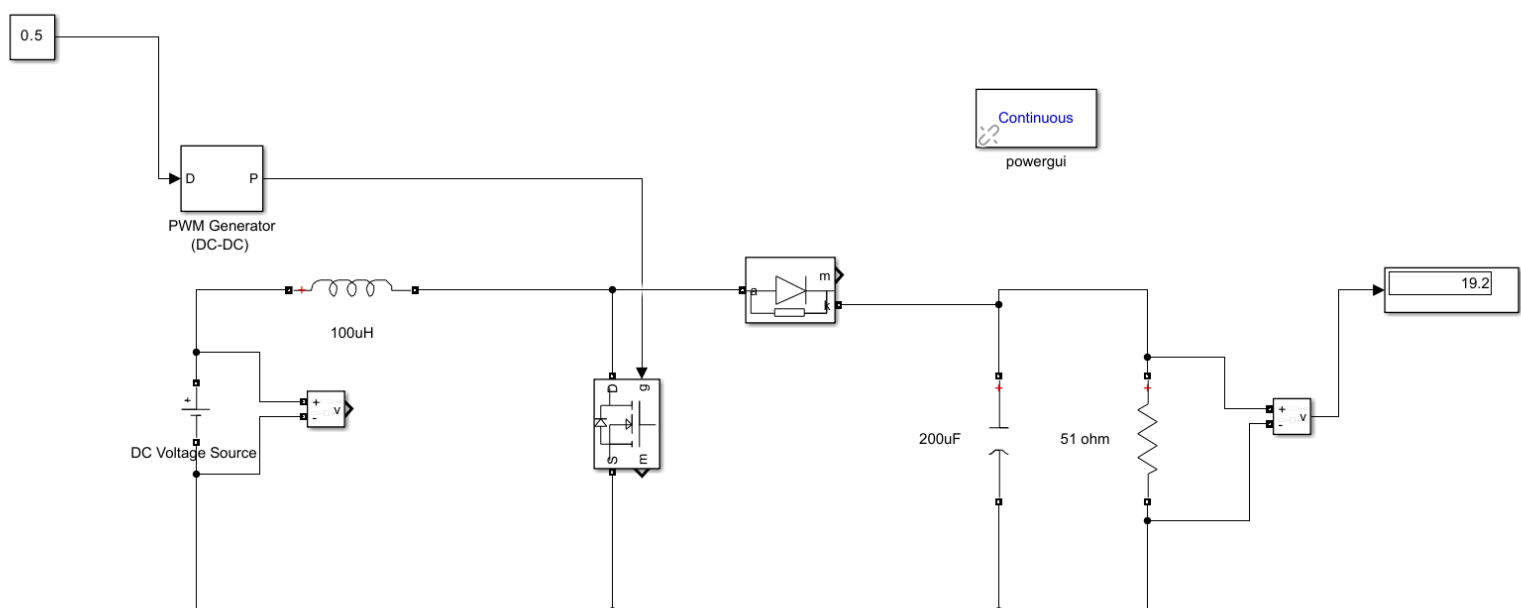
- Output voltage should stabilize near 20 V
- Output voltage ripple should be very small (≈ 18 mV)
- Inductor current should show a triangular waveform with approximately 1 A ripple
- Average inductor current should be approximately 0.784 A

Step 12: Data Export and Analysis

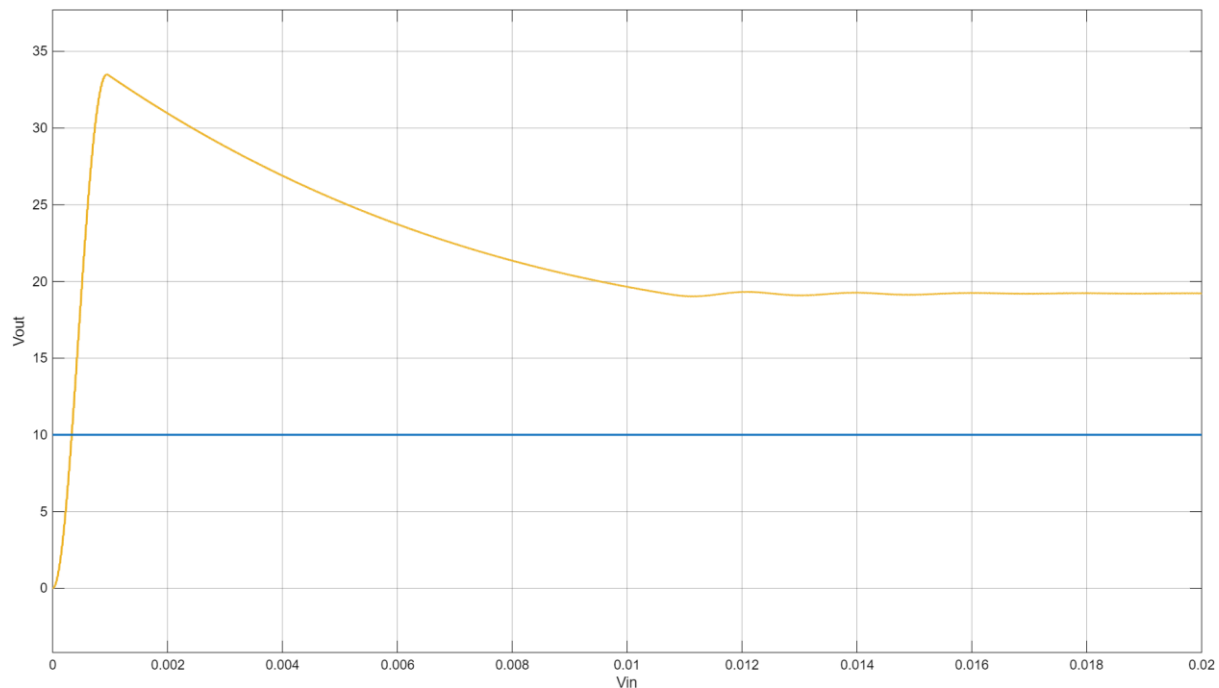
1. Double-click scope windows to export waveform data
2. Use To Workspace blocks to save variables for further analysis
3. Create plots in MATLAB to overlay results with theoretical calculations

3.5 Output Waveforms

Circuit Connection:



Output:



BUCK CONVERTER DESIGN

The buck converter is a fundamental step-down DC-DC converter topology. This section presents the working principle, mathematical foundations, design methodology, and Simulink implementation for a 10 V to 5 V converter at 50 kHz switching frequency.

4.1 Working Principle

A buck converter reduces input voltage to a lower output voltage using a controlled switch, freewheeling diode, inductor, capacitor, and load.

Switch ON Interval: Input voltage applied to inductor; diode reverse-biased; energy stored

Switch OFF Interval: Inductor releases energy through diode; output voltage maintained by capacitor

The average output voltage is directly:

$$V_{out} = D \times V_{in}$$

4.2 Important Buck Converter Formulae

The following relationships apply for ideal continuous conduction mode (CCM) operation:

Output Voltage Relation

$$V_{out} = D \times V_{in}$$

where D is the duty ratio (fraction of time switch remains ON).

Duty Ratio Relation

$$D = \frac{V_{out}}{V_{in}}$$

Inductor Ripple Current

$$\Delta I_L = \frac{V_{in} \cdot (1 - D)}{L \cdot f_s}$$

where:

- V_{in} = input voltage
- D = duty ratio
- L = inductor value
- f_s = switching frequency

Rearranging for inductor design:

$$L = L = \frac{(V_{in} - V_{out}) \times D}{\Delta I_L \times f_s}$$

Output Voltage Ripple

$$\Delta V_o = \frac{I_{out} \cdot D}{C \cdot f_s}$$

Rearranging for capacitor design:

$$C = \frac{I_{out} \cdot D}{\Delta V_o \cdot f_s}$$

Critical Inductance (CCM Boundary)

$$L_{crit} = \frac{(1 - D)R}{2f_s}$$

For continuous conduction mode, the selected inductor must satisfy: $L > L_{crit}$

Minimum Capacitance for Ripple Specification

$$C_{min} = \frac{I_{out} \cdot D}{\Delta V_o \cdot f_s}$$

4.3 Numerical Design for Buck Converter (Design-Based Component Selection)

Given: $V_{in} = 10 \text{ V}$, $D = 0.5$, $f_s = 50 \text{ kHz}$, $R = 51 \Omega$

Step 1-2 Calculations:

- $V_{out} = 0.5 \times 10 = 5 \text{ V}$
- $I_{out} = 5/51 = 0.098 \text{ A}$

Step 3 – Inductor Design:

- Assume $\Delta I_L = 30 \text{ mA}$
- $L = (5 \times 0.5)/(0.03 \times 50e3) = 1.67 \text{ mH}$
- Select: $L = 2.2 \text{ mH}$ (standard value with margin)

Step 4 – Capacitor Design:

- Assume $\Delta V_o = 10 \text{ mV}$
- $C = (0.098 \times 0.5)/(0.01 \times 50e3) \approx 98 \mu\text{F}$

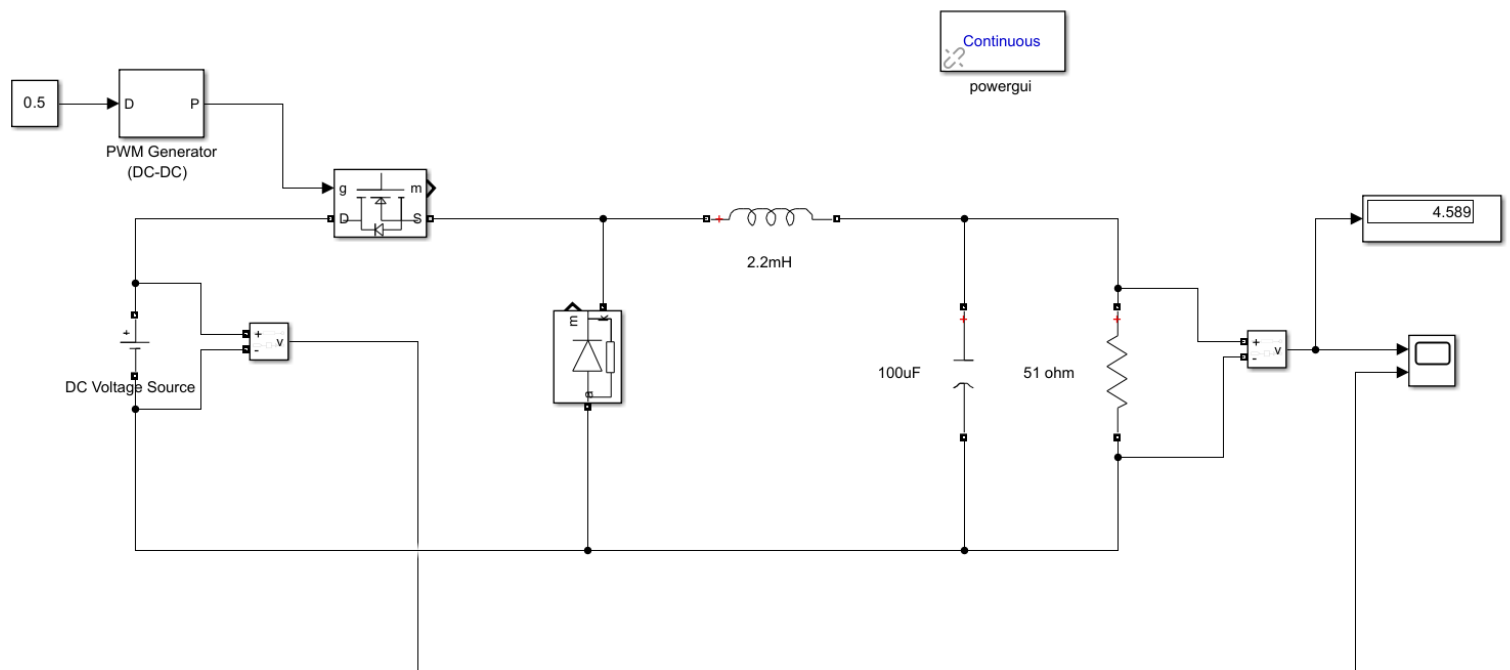
- Select: $C = 100\ \mu\text{F}$ (standard value)

4.4 Simulation Steps in Simulink

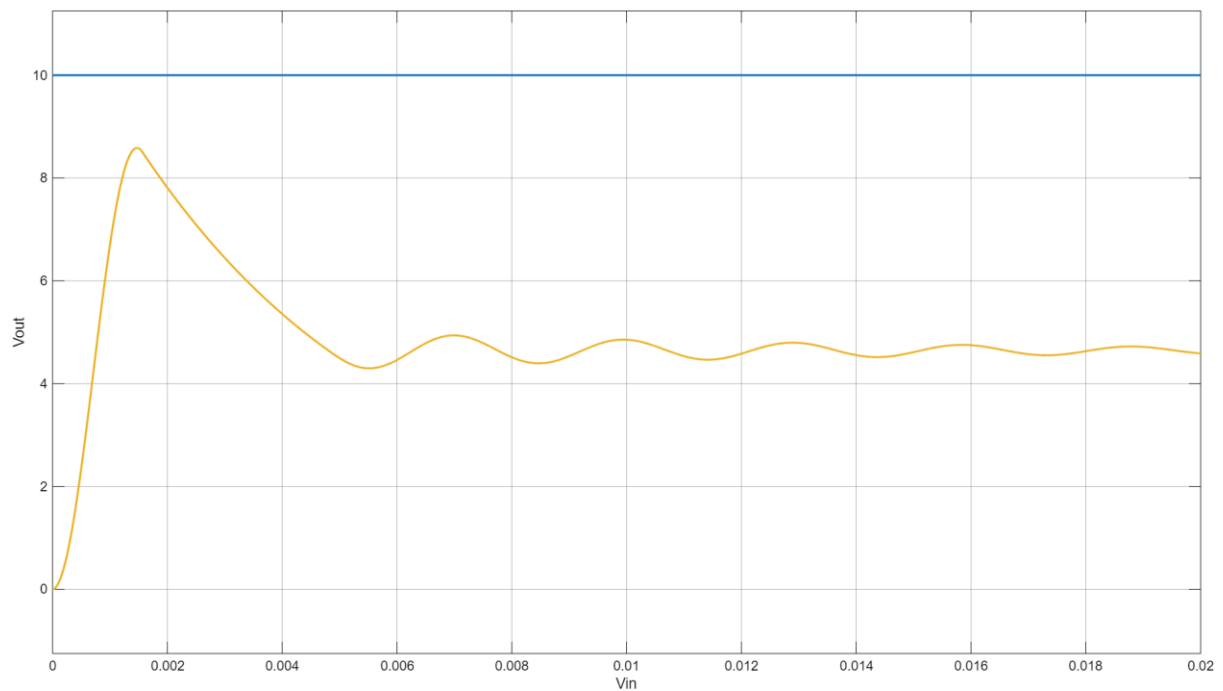
1. Create model: Buck_Converter.slx
2. Add DC source (10 V), switch, diode, inductor (2.2 mH), capacitor (100 μF), load (51 Ω)
3. Configure PWM: 50 kHz, 50% duty cycle
4. Add voltage and current measurement blocks
5. Set solver: Step size 1e-6 s, Stop time 20 ms
6. Run simulation
7. Verify output $\approx 5\ \text{V}$ with minimal ripple

4.4 Output Waveforms

Circuit Connection:



Output :



COMPARATIVE ANALYSIS & THROW-BACK

5.1 Comparative Analysis of Converters

Feature	Boost Converter	Buck Converter
Function	Step-up voltage	Step-down voltage
Output relation	$V_{out} = V_{in}/(1 - D)$	$V_{out} = D \cdot V_{in}$
Duty ratio	0.5	0.5
Input voltage	10 V	10 V
Output voltage	20 V	5 V
Output current	0.392 A	0.098 A
Inductor value	100 μ H	2.2 mH
Capacitor value	220 μ F	100 μ F
Inductor ripple	1 A	30 mA
Voltage ripple	18 mV	10 mV
Critical inductance	64 μ H	0.255 mH
Typical application	Raise DC bus	Reduce DC bus

Key Insight: Boost is chosen when source voltage is too low; buck when it's too high. The design methodology remains consistent: assume ripple → calculate components → verify CCM.

5.2 Gate Driver Board and PWM Control

Purpose of Gate Driver:

Gate drivers provide:

- Proper voltage level (typically 12-15 V for MOSFETs)
- High peak current for fast switching (reduces losses)
- Electrical isolation (if required)
- Protection against noise and transients

PWM Modulation:

Pulse Width Modulation controls output voltage by varying switch ON time:

$$V_{out,avg} = f(D, V_{in})$$

where duty ratio $D = \text{ON time} / \text{Total period}$

At 50 kHz, $T_{period} = 20$ microseconds

Why Gate Drivers Are Essential

- Control circuits cannot directly drive power MOSFETs (insufficient current/voltage)
- Fast switching reduces switching losses from $P = f \cdot C_w \cdot V^2$
- Proper gate drive enables soft switching techniques
- Isolation protects low-power controller from high-power stage

5.3 Brief Overview of the 5-Level Multilevel Inverter

Multilevel Inverter Concept

A multilevel inverter generates AC waveforms using more than two discrete voltage levels. This reduces harmonic content and improves waveform fidelity compared to traditional two-level inverters.

Advantages of 5-Level Topology

- THD Reduction: Lower total harmonic distortion (typically 30-50% reduction vs. 2-level)
- Reduced dv/dt Stress: Slower voltage transitions reduce EMI and bearing currents
- Better Efficiency: Lower switching frequency per device while maintaining output quality
- Reduced Filtering: Less passive filtering required due to inherently lower harmonics

Five Voltage Levels

A 5-level inverter synthesizes:

$$\{-2V_{dc}, -V_{dc}, 0, +V_{dc}, +2V_{dc}\}$$

This creates a staircase approximation to sine wave with better fidelity than 3-level ($\{-V_{dc}, 0, +V_{dc}\}$)

Common Topologies

Topology	Characteristics
Cascaded H-Bridge	Modular, scalable, easy to understand
Flying Capacitor	Reduced device count, voltage balancing required
Diode-Clamped	Mid-point neutral, clamping diodes

For educational use, cascaded H-bridge is most common.

Switching Control Methods

- Sinusoidal PWM (SPWM): Reference sine wave compared with triangular carrier
- Phase-Shifted PWM: Different carriers for different modules
- Space Vector PWM: Optimal switching for minimum harmonics
- Selective Harmonic Elimination: Targeted harmonic cancellation

DC Link Requirements

For practical systems:

- Target AC output: 230 V RMS (about 325 V peak)
- Required DC link: 400-500 V (accounting for modulation losses)
- Our 10 V $\rightarrow$ 20 V boost is a low-voltage educational example

Key Design Parameters

Parameter	Value
Output frequency	50-60 Hz (AC mains)
Switching frequency	1-10 kHz (lower than DC-DC)
Modulation index	$M = 0.8-0.95$
Output power	Watts to Megawatts (application dependent)
❖ The 5-level inverter stage represents the AC conversion portion of the power chain, demonstrating how regulated DC voltage is converted to clean AC.	

CONCLUSION AND REFERENCES

6.1 Conclusion

This project presented the design and simulation of boost and buck DC-DC converters operating from a 10 V DC source at a switching frequency of 50 kHz. The boost converter successfully stepped the voltage up to 20 V, while the buck converter reduced the voltage to 5 V. The inductor and capacitor values were selected using design equations based on ripple current, ripple voltage, and continuous conduction mode conditions.

The project also included a conceptual study of the five-level cascaded H-bridge multilevel inverter using SPWM technique. The multilevel inverter topology provides improved output waveform quality, reduced harmonic distortion, and better efficiency compared to conventional two-level inverters. Simulation results verified the theoretical analysis and demonstrated proper operation of the converters. Overall, the project helped in understanding the practical design methodology of DC-DC converters, PWM control techniques, and multilevel inverter systems used in modern power electronic applications.

6.2 References

Textbook:

Ned Mohan, Tore M. Underland, William P. Robbins, Power Electronics: Converters, Applications, and Design, 3rd Edition, Wiley, 2002.

- ❖ Classic treatment with emphasis on design calculations and CCM/DCM analysis

Software Documentation

MathWorks, MATLAB and Simulink Documentation [Online]

- ❖ Simscape Electrical, Power Systems Block set, and simulation solver documentation

MathWorks, State-Space Representation of DC-DC Converters, Application Note, 2019.

- ❖ Modelling and simulation guidelines for converter design